

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of MSc Physics Examination November 2019

PS904 Numerical Analysis

Date: 13-11-2019, Day: Wednesday Time: 01.30 PM To 01.50 PM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q-1 Choose the correct answer for the following questions [10]

1. The value of $\int_0^2 x \, dx$ by using Simpson's 1/3 rule most nearly is 1
A. 0.3333 B. 1 C. 0.6666 D. 2
2. The atomic weight of silicon is 28.0855. Round this to 3 significant figures. 1
A. 28.1 B. 28.0
C. 28.09 D. 28.086
3. If the exact root is 21.32 and approximate root using Simple fixed point method is 21.23 then true fractional relative error is 1
A. 1.04% B. 4.2%
C. 0.09 D. 0.0042
4. In which of the following method, we approximate the curve of solution by the tangent in each interval. 1
A. Euler's method B. Newton's method
C. Cubic Spline D. Simpson's method

5. Which of the following statements applies to the bisection method used for finding roots of functions? 1
- Converges within a few iterations
 - Guaranteed to work for all continuous functions
 - Is faster than the Newton-Raphson method
 - Derivative of the function should be non-zero
6. Which of these statement(s) is/are correct? 1
- As step size/width (h) increases, accuracy of integration increases
 - Newton divided difference method yields the same result as Lagrange interpolation
 - You require n data points to fit the n^{th} order of polynomial.
 - None of the above
7. In which of the following methods proper choice of initial value is very important? 1
- Bisection method
 - False position
 - Newton-Raphson
 - Simple fixed point iteration
8. Match the following: 1
- | | |
|-------------------|---|
| a. Secant method | 1. Integration |
| b. Euler's method | 2. Root finding |
| c. Lagrange | 3. Ordinary Differential Equations solver |
| d. Trapezoidal | 4. Interpolation |
- The correct sequence is
- | | |
|----------------|----------------|
| A. a2-b4-c1-d3 | B. a3-b2-c1-d4 |
| C. a1-b4-c2-d3 | D. a2-b3-c4-d1 |
9. The highest order of polynomial integrand for which Simpson's 1/3 rule of integration is exact is 1
- | | | | |
|----------|-----------|----------|-----------|
| A. First | B. Second | C. Third | D. Fourth |
|----------|-----------|----------|-----------|
10. Given best fitting data to by least squares requires minimization of 1
- $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$, best fitting data to $y = f(x)$ by least squares requires minimization of
- $\sum_{i=1}^n [y_i - f(x_i)]$
 - $\sum_{i=1}^n |y_i - f(x_i)|$
 - $\sum_{i=1}^n [y_i - f(x_i)]^2$
 - $\sum_{i=1}^n [y_i - \bar{y}]^2$, where $\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$

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PS904 Numerical Analysis

Date: 13-11-2019, Day: Wednesday Time: 01.50 PM To 03.30 PM Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.

SECTION – I

Q - II Answer the following questions as directed

10

1. $f(x) = -2x^6 - 1.5x^4 + 10x + 2$

4

Use a root location technique to determine the maximum of the above function. Perform iteration until the approximate relative error falls below 5%. If you use bracketing method, use: $x_l = 0$ and $x_u = 1$. If you use Newton-Raphson method, use an initial guess of $x_i = 1$. For secant method, $x_{i-1} = 0$ and $x_i = 1$. Assuming that convergence is not an issue, choose the technique that is best suited to this problem. Justify your choice.

2.

x	1	2	3	5
$f(x)$	3	6	19	99

3

Calculate $f(4)$ using Newton's or Lagrange interpolating polynomials of order 1 and 2.

3. (a) Solve the integral using single segment and three segment trapezoidal rule $\int_{0.2}^{2.2} xe^x dx$.
(b) Write formula for multiple application of Simpson's 1/3 rule.

3

SECTION – II

Q-III Answer the following questions as directed.

15

1. Use any bracketing method to determine the drag coefficient c needed for a parachutist of mass $m = 68.1$ kg to have a velocity v of 40 m/s after free falling for time $t = 10$ s.

$$f(c) = \frac{9.8m}{c} (1 - e^{-(ct/m)}) - v$$

The root may be located between 12 and 16. Compute the root until ϵ_a is less than 2%

2. Fit a straight line to the x and y values given in the following Table:

4

x	1	2	3	4	5	6	7
y	0.5	2.5	2.0	4.0	3.5	6.0	5.5

OR

OR

A torsion spring of a mousetrap is twisted through an angle of 180° . The torque vs. angle data is given below. 4

Torsion, T (N-m)	0.110	0.189	0.230	0.250
Angle, θ (rad)	0.10	0.50	1.1	1.5

If the relationship between the torque and the angle is : $T = a_0 + a_1\theta$. Find a_0 and a_1

3. The deflection y of the top of a sailboat mast is:

4

$$y = \frac{FL^4}{8EI}$$

Where, $F = 750 \text{ N/m}$

$\Delta F = 30 \text{ N/m}$

$L = 9 \text{ m}$

$\Delta L = 0.03 \text{ m}$

$E = 7.5 \times 10^9 \text{ N/m}^2$

$\Delta E = 5 \times 10^7 \text{ N/m}^2$

$I = 5 \times 10^{-4} \text{ m}^4$

$\Delta I = 5 \times 10^{-6} \text{ m}^4$

Estimate the error in y .

4. Some rangers estimate that in their park the deer population $p(t)$, in thousands, is governed by the logistic differential equation $dp/dt = 0.1p(1 - p)$ and that the current size of the population is 0.4, that is $p(0) = 0.4$. Using Euler's Method with $\Delta t = 5$, what will be the predicted deer population (in thousands) 15 years from now? 4

OR

Apply Euler's method to approximate the solution of the initial value problem

4

$$\frac{dy}{dx} = x + \frac{1}{5}y, \quad y(0) = -3 \text{ with step size } h=0.2 \text{ on the interval } [0,1].$$